

Project Assignment Dillema

A company is distributing software development projects among a group of developers. Each developer has a preference for either a backend or frontend project, denoted by 0 and 1 respectively.

The company has a stack of projects to be assigned in a fixed order, where:

- `projects[i]` represents the type of the i -th project in the stack ($i = 0$ is the topmost project).
- `developers[j]` represents the preference of the j -th developer in the initial queue ($j = 0$ is the first developer in line).

At each step:

- If the first developer in the queue prefers the topmost project, they take it from the stack and leave the queue.
- Otherwise, they reject the project and move to the end of the queue.

This process continues until no developers in the queue want to take the project at the top of the stack, and further assignments are impossible. Your task is to determine how many developers will be unable to get a project after the process terminates.

Input Format

- The first line contains an integer n - the number of developers and projects (both are equal).
- The second line contains an array of n integers (developers) where each value is 0 (prefers backend) or 1 (prefers frontend).
- The third line contains an array of n integers (projects) representing the available projects in stack order.

Output Format

Return a single integer representing the number of developers who cannot be assigned a project.

Constraints

- $1 \leq n \leq 100$
- Implement your algorithm in C using Single Linked List to make the Stack and the Queue.

Sample Input 1 (standard input)

```
4
1 1 0 0
0 0 1 1
```

Sample Output 1 (standard output)

```
0
```

Explanation 1

```
Developer 1 (prefers frontend) rejects backend project, moves to queue end.
Current queue: 1 0 0 1. Current stack: 0 0 1 1
Developer 2 (prefers frontend) rejects backend project, moves to queue end.
Current queue: 0 0 1 1. Current stack: 0 0 1 1
Developer 3 (prefers backend) accepts backend project, Leaves queue.
Current queue: 0 1 1. Current stack: 0 1 1
Developer 4 (prefers backend) accepts backend project, Leaves queue.
Current queue: 1 1. Current stack: 1 1
Developer 1 (prefers frontend) accepts frontend project, Leaves queue.
Current queue: 1. Current stack: 1
Developer 2 (prefers frontend) accepts frontend project, Leaves queue.
Current queue: NULL. Current stack: NULL
```

Sample Input 2 (standard input)

```
6
1 1 1 0 0 1
1 0 0 0 1 1
```

Sample Output 2 (standard output)

```
3
```